

## Nikita Safonkin

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**Date of birth** 28 September 1996  
**Place of birth** Russia, Moscow  
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**Personal website** <https://www.nikita-safonkin.com>  
**Employment**  
2023 – present Postdoctoral researcher at Leipzig University

### Education

2017 Lomonosov Moscow State University, Faculty of Physics, Department of Mathematics, B.Sc. in Physics  
2019 National Research University Higher School of Economics, Faculty of Mathematics, M.Sc.; Skolkovo Institute of Science and Technology, M.Sc.  
2023 University of Reims Champagne-Ardenne, Ph.D.

### PhD degree

2023 *Harmonic functions on branching graphs*

Advisor Grigori Olshanski

### Papers

2026 N.Safonkin, "Trace radicals and cocenters of free products", [arXiv:2607.24697](#).  
2026 M.Fairon, N.Safonkin, "Double Transposed Poisson Algebras", [arXiv:2607.01066](#).  
2026 N.Safonkin, "Coupled double Poisson brackets", Journal of Algebra 708, 452–491, [link](#) and [arXiv:2605.17696](#).  
2025 N.Safonkin, "What is a double star-product?", [arXiv:2506.00699](#).  
2023 G.Olshanski, N.Safonkin, "Double Poisson brackets and involutive representation spaces", Letters in Mathematical Physics 114, 33, [arXiv:2310.01086](#).  
2023 G.Olshanski, N.Safonkin, "Remarks on Yangian-type algebras and double Poisson brackets", Functional Analysis and Its Applications 57, 326–336, [arXiv:2308.13325](#).  
2022 P. Nikitin, N. Safonkin, "Semifinite harmonic functions on the direct product of graded graphs", Zapiski Nauchnykh Seminarov POMI, Volume 517, 125–150, [arXiv:2209.05901](#).  
2022 N.Safonkin, "Semifinite harmonic functions on the zigzag graph", Functional Analysis and Its Applications, 56:3, 52–74, [link](#) and [arXiv:2110.01508](#).  
2022 N.Safonkin, "Semifinite Harmonic Functions on Branching Graphs", Journal of Mathematical Sciences (New York), 261, 669–686, [link](#) and [arXiv:2108.07850](#).  
2021 N.Safonkin, "Semifinite Harmonic Functions on the Gnedin-Kingman Graph", Journal of Mathematical Sciences (New York) 255, 132–142, [link](#) and [arXiv:2103.02257](#).  
2018 P. Bibikov, N. Safonkin, "Classification of Certain Class of Ordinary Differential Equations of the First Order", Russian Mathematics 62, [link](#).

### Research visits

March – June 2023 Weizmann Institute of Science, Rehovot, Israel.  
December 2021 Leonhard Euler International Mathematical Institute in Saint Petersburg.

### Awards

2021 IUM-Simons foundation scholarship.

### Teaching

2021 Teaching assistant at the Independent University of Moscow.  
2022 Algebra-1 class tutor at the Higher School of Economics, Faculty of Mathematics.

### Talks

- 2025 *Deformation quantization of double Poisson algebras*, BIMSIA Integrable Systems Seminar, [link](#).
- 2025 *What is a double star-product?*, Deformation quantization and quantum groups seminar, The Independent University of Moscow [link](#).
- 2025 *What is a double star-product?*, X International Workshop on Non-Associative Algebras in Dijon [link](#).
- 2025 *What is a double star-product?*, Séminaire Physique mathématique ICJ, Institut Camille Jordan, [link](#).
- 2025 *What is a double star-product?*, European Non-Associative Algebra Seminar, [link](#).
- 2024 *Double Poisson brackets and involutive representation spaces*, Geometry seminar, Max Planck Institute for Mathematics in the Sciences, Leipzig, [link](#).
- 2024 *Double Poisson brackets and involutive representation spaces*, VIII International Workshop on Non-Associative Algebras in Linköping, [link](#).
- 2023 *Yangian-type algebras and double Poisson brackets*, October 3, Deformation quantization and quantum groups seminar, IUM, [link](#).
- 2023 *Yangian-type algebras and double Poisson brackets*, September 26, BIMSIA Integrable Systems Seminar, [link](#).
- 2023 *Yangian-type algebras and double Poisson brackets*, July 11, University of Reims Champagne-Ardenne, [link](#).
- 2023 *Semifinite harmonic functions on the zigzag graph*, Les Probab du vendredi, LPSM, Sorbonne University, January 27, [link](#).
- 2022 *Graded graphs and related topics*, Working Seminar on Mathematical Physics of HSE and Skoltech CAS, September 21, [link](#).
- 2021 *Wassermann's method for the Young and zigzag graphs*, St. Petersburg Seminar on Representation Theory and Dynamical Systems, PDMI, December 22, [link](#).
- 2021 *Semifinite harmonic functions on the Young and zigzag graphs*, St. Petersburg Seminar on Representation Theory and Dynamical Systems, PDMI, December 8, [link](#).
- 2021 Several talks at [Representations and Probability seminar](#):
- *Semifinite harmonic functions*, March 29, [video](#).
  - *Harmonic functions on the zigzag graph*, March 22, [video](#).
  - *The zigzag graph and the path space of a branching graph*, March 15, [video](#).
  - *Harmonic functions on the Gnedin-Kingman graph and Kerov's construction*, March 1, [video](#).
  - *Monomial quasisymmetric functions and the Gnedin-Kingman graph*, February 18, [video](#).
  - *Harmonic functions on the Pascal, Kingman and Young graphs 2*, February 11, [video](#).
  - *Harmonic functions on the Pascal, Kingman and Young graphs 1*, February 4, [video](#).
- 2021 *The algebra of quasisymmetric functions and the Gnedin-Kingman graph*, [Seminar on Lie algebras and applications](#), March 2.
- 2021 *Macdonald polynomials, Kerov's construction, and Matveev's theorem*, [Working Seminar on Mathematical Physics of HSE University and Skoltech Center for Advanced Studies](#), February 24, [video](#).

- 2021 *Harmonic functions on the Kingman and Young graphs*, [Working Seminar on Mathematical Physics of HSE University and Skoltech Center for Advanced Studies](#), February 10, [video](#).
- 2020 *Semifinite harmonic functions on the Gnedin-Kingman graph*, Central and invariant measures and applications, August 17-21, 2020, Euler International Mathematical Institute, St. Petersburg, Russia.
- 2019 *Unitary representations of the quantized algebra  $U_q(\mathfrak{u}(m, n))$* , Research Seminar of the Master's Program in Mathematics and Mathematical Physics at the Higher School of Economics.